Plural, paucal, or both:
A reanalysis of the Russian quantified construction

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Abstract In this paper, I argue that neither nonagreeing plural accounts nor agreeing nonplural accounts of the Russian paucal construction are satisfactory in addressing some long-standing puzzles around this topic. As an alternative, I propose a fully agreeing approach, where the quantified NP as a whole and nodes thereunder may be uniformly specified for number at the morphology–syntax interface as [–pc, +pl] ("plural exclusive"), [+pc, –pl] ("paucal exclusive"), or [+pc, +pl] ("paucal plural"), among others. With this unified analysis for both quantified and unquantified NPs, the existing puzzles can be solved in a concise and consistent way.

1 Russian quantified construction: basic facts

The Russian quantified construction (henceforth QNP) is known for its puzzling morphosyntactic behaviors. In traditional descriptions, the following overriding rules (to be revised) are needed for QNPs involving **paucal** or **multal** numerals in **direct** cases, rather than simply using the otherwise expected nominative or accusative plural:

- The "head noun" takes the **genitive** plural when the numeral is multal, or the **genitive singular** when the numeral is paucal.
- Any "adjective within the NP" takes the **genitive** plural, but can also take the nominative plural when the noun is feminine and the numeral is paucal.

In the context of the Russian QNP, we say that:

- Numerals 1, 21(!), … are lexically **singular**; 2, 3, 4, 22, … are lexically **paucal**; 5~20, 25, … are lexically **multal**.[1]
- The (Q)NP bears a **direct** case when it bears the nominative case or the accusative case, unless the accusative case is syncretized to the genitive case[2]; otherwise it bears an **oblique** case.

Some applications of these rules are exemplified in (1):

(1)	a.	*dvumja/pjat'ju*	*bol'šimi*	*knigami/parkami*	"by {2, 5} large books/parks" (INS)
		two/five.INS	large.PL.INS	book(F)/park(M).PL.INS	
	b.	*pjat'*	*bol'šix*	*knig/parkov*	"5 large books/parks" (NOM)
		five.NOM	large.PL.**GEN**	book(F)/park(M).PL.**GEN**	
	c.	*dva*	*bol'šix*	*parka*	"2 large parks" (NOM)
		two.NOM.M/N	large.PL.**GEN**	park(M).**SG.GEN**	
	d.	*dve*	*bol'šie~bol'šix*	*knigi*	"2 large books" (NOM)
		two.NOM.F	large.PL.NOM~**GEN**	book(F).**SG.GEN**	

- (1a) is an oblique (instrumental) paucal/multal phrase, where the adjective and the noun agree in case and number with the whole phrase.
- (1b) is a direct multal phrase, where the adjective and the noun both take the **genitive** plural, instead of the otherwise expected nominative plural.

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[1] As a definition, a nonnegative numeral x ($x \bmod 1000 \neq 0$) taking the form of an integer is lexically classified as: **multal**, when (($x \bmod 10 \in \{0, 5, 6, 7, 8, 9\}$) or ($x \bmod 100 \in \{11, 12, 13, 14\}$)); or otherwise **paucal**, when $x \bmod 10 \in \{2, 3, 4\}$; or otherwise **singular**. Fractions in form, integers that are multiples of 1000, and the so-called "collective numerals" are excluded from our discussion.

[2] The accusative phrase is subject to the syncretism condition if it is surficially nonsingular animate or masculine singular animate. A surficially nonsingular phrase is considered surficially animate only when (a) the noun is lexically animate, and (b) the numeral is "collective" or less than 5, and (c) the phrase is not governed by a "quantificational" verb or preposition, …. A precise formulation of the surface animacy condition is beyond the scope of this paper. See Mel'čuk (1980; 1985: 199-203) for more.

- (1c) is a direct paucal phrase headed by a nonfeminine noun, where the adjective takes the **genitive** plural, but the noun takes the **genitive singular**.
- (1d) is a direct paucal phrase headed by a feminine noun, where the adjective can take either the **nominative or genitive** plural, but the noun can only take the **genitive singular**.

The overriding rules listed at the beginning of this section need one important revision. Nouns converted from adjectives, like *stolovaja* ("canteen", from *stolovaja* "related to dinner" (feminine)), follow the same pattern as adjectives in taking GEN.PL when serving as the head noun of a paucal phrase in direct cases. The declension type of the syntactic nodes is the decisive factor in conditioning the choice of wordforms in a paucal phrase, not the part of speech.

We have so far gone through the most important facts about the Russian QNP. In § 2, I discuss in turn the nonagreeing and nonplural analyses of the paucal construction respectively and how they fail to accommodate certain puzzles. § 3 presents the reanalysis and explains how it cuts the Gordian knot.

2 Existing analyses and problems

2.1 Nonagreeing analyses

The superficial nonagreement within the direct paucal/multal constructions, and the apparent discrepancy between the nonagreeing pattern in direct cases and the agreeing pattern in oblique cases, call for theoretical accounts. Traditional grammars generally give no explanation for this, with some acknowledging its status as a historical relic and its irrationality within modern Russian grammar (Vinogradov 1986: 246). More recently, there have been a number of theoretically oriented analyses attempting to account for the case and number nonagreements involved here, mostly by attributing the case nonagreement to the numeral's government of the genitive case over the noun. In order for the government to apply only to direct cases and not to disrupt fully agreeing oblique cases, they have variously resorted to the syntactic case hierarchy (Babby 1987) and difference in headedness (Neidle 1988), among others. Despite these efforts, Babby and Neidle have left unexplained the number nonagreement between the noun and the adjective in the direct paucal. Pesetsky's (2013) theory, developed within his theoretical framework, approaches the case nonagreement by assuming nouns as underlyingly genitive unless overridden, and the number nonagreement by maintaining nouns in the direct paucal as numberless and yet to be merged with the number feature before being modified by plural adjectives.

There are, however, major difficulties with these nonagreeing analyses, particularly in addressing the Ivanov(a) puzzle and the mountain puzzle. I defer the mountain puzzle to § 2.5, and introduce the Ivanov(a) puzzle first.

2.2 The Ivanov(a) puzzle: *dva Ivanova* but not *dve *Ivanovoj*

The *Ivanov(a)–Nikitin(a)* declension, comprised of surnames ending in *-in(a)* and *-ov(a)* (and common nouns converted therefrom), is challenging to nonagreeing analyses. Its peculiarity in the direct paucal, which I will call "the Ivanov(a) puzzle", has been noted in various works in the literature (Tolbert 1974: 29; Meľčuk 1985: 177; Billings 1995: 57; Franks 1995: 52; Pereltsvaig 2009).

This declension is traditionally called a mixed declension, as it conflates two declensions into one (see (2)): it takes all cells except M.SG.GEN/DAT/PREP from the possessive adjectival declension (shaded blue and purple), and takes all direct case cells and M.SG.GEN/DAT/PREP cells from the masculine nominal declension (shaded red and purple). Here direct cases can be construed as belonging to either declension (shaded purple).

(2) *Ivanóv(a)* "Ivanov(a)" (Noun.ANIM)

	SG.F	SG.M	PL
NOM	*Ivanóva*	*Ivanóv*	***Ivanóvy***
ACC	*Ivanóvu*		
GEN	*Ivanóvoj*	***Ivanóva***	***Ivanóvyx***
PREP		*Ivanóve*	
DAT		*Ivanóvu*	*Ivanóvym*
INS	*Ivanóvoj(u)*	*Ivanóvym*	*Ivanóvymi*

For "3 Ivanov(a)s" in direct cases, there are three possibilities in all (see Pereltsvaig 2009):[3]

- *tri Ivanovyx* (= GEN.PL): "3 Ivanov(a)s", any single or mixed gender;
- *tri Ivanovy* (= NOM.PL): "3 Ivanovas", all female;
- *tri Ivanova* (= GEN.SG.M): "3 Ivanovs", all male.

Ivanovyx and *Ivanovy* pose no problem here as they are regularly introduced by the nonagreement rule for adjectival declension nodes. However, while *Ivanova* (= GEN.SG.M) seems to be introduced by the nonagreement rule for nominal declension nodes, the mystery is why only the masculine GEN.SG is grammatical (*tri Ivanova*), but not its feminine counterpart (*tri *Ivanovoj*). One may be tempted to split *Ivanov* and *Ivanova* into two different lexemes, an adjectival *Ivanova* and a doubly classified nominal~adjectival *Ivanov*, to account for the difference (as implicitly assumed in Pesetsky's (2013: 122) approach to the puzzle). Unfortunately, from the mixed-gender expression *tri Ivanovyx* it is evident that *Ivanov* and *Ivanova* belong to the same lexeme. It is thus hard to imagine why this minor declension class, whose members are proper names that rarely occur in QNPs, could have stood up to the pressure of grammatical leveling.

This puzzle, viewed from a broader perspective, stems from the difficulty in drawing a clear-cut distinction between "nouns" and "adjectives" in the syntax to account for the number nonagreement in the direct paucal. In a theory that crucially demands such a side-taking in the syntax, one can arguably treat head nouns of the adjectival declensions (like *stolovaja*) formally as a dependent adjective plus a zero head noun, but as a consequence the *Ivanov(a)* family is torn apart—while *Ivanova* seems to have readily taken the side of the adjective, *Ivanov* is still sitting on the fence.

2.3 Nonplural analyses

In the face of the difficulties encountered in nonagreeing analyses, agreeing analyses where nodes within a QNP share the same morphosyntactic labels have been put forward. This is achieved by postulating additional labels compared with the traditional analysis, say, by establishing "paucal" as a formal morphosyntactic label in Russian grammar rather than just a typological term used for cross-linguistic comparison. Thus the paucal is recognized as a third number in addition to the singular and the plural, rather than being subsumed under the plural.

Bailyn & Nevins' (2008) work is a representative analysis along this line, where they argue that the noun and the adjective of a QNP in the direct paucal are morphosyntactically nominative or accusative in case and paucal in number. In this way, a fully agreeing pattern like (3) can be established for all paucal phrases, and the aforementioned troubles in delimiting nouns and adjectives are avoided:

(3) "2 large books" (NOM) (surface analysis ⇒ Bailyn & Nevins' analysis)

a.	*dve*	*bol'šie*	*knigi*
	two.NOM.F	large.PL.NOM⇒PC.NOM.F	book(F).SG.GEN⇒PC.NOM
b.	*dve*	*bol'šix*	*knigi*
	two.NOM.F	large.PL.GEN⇒PC.NOM[4]	book(F).SG.GEN⇒PC.NOM

Yadroff's (1999), Rakhlin's (2003), and Pereltsvaig's (2009) proposals are very close to Bailyn & Nevins', except that the direct paucal is recognized morphosyntactically not as the nominative/accusative case but as a separate case called the "adpaucal case", the "genitive of quantification", or the "quantificational case" in the paucal number. These differences don't concern us here. Unfortunately, these nonplural analyses all fail to account for Suprun's puzzle and the mountain puzzle, which I shortly turn to.

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[3] Disregarding the expression with a collective numeral, *troe* (collective) *Ivanovyx* (GEN.PL) ("3 Ivanov(a)s", any single or mixed gender).

[4] Bailyn & Nevins' original presentation of their analysis, where *bol'šix* in (3b) would be glossed as PC.NOM.M/N, would judge (3b) as ungrammatical due to failed agreement in gender, thus providing no room for the adjective variation in the direct paucal. This, of course, can be trivially fixed by identifying the adjective form syncretized to the genitive plural (*bol'šix*) as simply PC.NOM unspecified for gender.

2.4 Suprun's puzzle

As mentioned above, the adjective in a feminine QNP in the nominative paucal can take a wordform syncretized to either NOM.PL or GEN.PL, and we know that there are usually fluctuations between plural agreement and singular nonagreement of the verb with a quantified subject. Suprun (1957: 76~77) notes an interesting interaction between these two variations: among the four potential possibilities of combination, only three are possible, ruling out a NOM.PL adjective in the subject QNP with a nonagreeing neuter singular predicate verb[5].

(4) "On the table lay four large books." (from Corbett 2006: 196)

a.	✗	*Na stole*	*ležalo*	*četyre*	*bol'šie*	*knigi.*
		on the table	lie.PST.**SG.N**	four.NOM	large.**PL.NOM**⇒PC.NOM.F?	book(F).SG.GEN
b.	✓	”	*ležali*	”	*bol'šie*	”
		”	lie.PST.**PL**	”	large.**PL.NOM**⇒PC.NOM.F?	”
c.	✓	”	*ležalo*	”	*bol'šix*	”
		”	lie.PST.**SG.N**	”	large.**PL.GEN**⇒PC.NOM?	”
d.	✓	”	*ležali*	”	*bol'šix*	”
		”	lie.PST.**PL**	”	large.**PL.GEN**⇒PC.NOM?	”

The problem is how this difference between PC.NOM.F and PC.NOM (or PC.NOM.M/N in Bailyn & Nevins' original presentation) can lead to the difference in grammaticality between (4a) and (4c).

2.5 The mountain puzzle

There is an oft-mentioned rule (Tolbert 1975: 26; Dunn 2009: 180; Wade 2010: 215; Pesetsky 2013: 123) concerning the use of adjectives in the direct paucal, which I will name after the example given in Rosenthal's rules (#193.2.3) as "the mountain rule": if GEN.SG and NOM.PL of a feminine noun differ (almost always in stress if they do), the modifying adjective can only take GEN.PL (but not NOM.PL) in the direct case of a paucal phrase. This rule, although no longer strictly obeyed in 21st-century Russian (Pereltsvaig 2009; Šaronov 2014: 143), is a reflection of what the language used to be back in the 20th century.

(5) 20th-c. Russian > 21st-c. Russian

a.	✓	*dve*	*bol'šie*	*knígi*	"2 large books" (NOM)
		two.NOM.F	large.PL.NOM	book(F).SG.GEN(=PL.NOM)	
b.	✓	”	*bol'šix*	*knígi*	”
		”	large.PL.GEN	book(F).SG.GEN(=PL.NOM)	
c.	✗>✓	”	*bol'šie*	*gorý*	"2 large mountains" (NOM)
		”	large.PL.**NOM**	mountain(F).SG.GEN(**≠PL.NOM**)	
d.	✓	”	*bol'šix*	*gorý*	”
		”	large.PL.GEN	mountain(F).SG.GEN(≠PL.NOM)	
e.	*	”	(*bol'šie*/*bol'šix*)	*góry*	”
		”	(large.PL.NOM/GEN)	mountain(F).**PL.NOM**(≠SG.GEN)	

As mentioned above, a feminine head noun with homophonous GEN.SG=NOM.PL allows the variation of the adjective in the direct paucal, as in (5a, b). In 20th-c. Russian, however, the NOM.PL option for the adjective was unacceptable (though no longer today) when the two wordforms were not homophonous and the noun clearly took the form of GEN.SG and not NOM.PL, as in (5c). It should be noted that the noun taking NOM.PL is unacceptable either, as in (5e).

In a nonagreeing approach, the homophony of GEN.SG and NOM.PL, being purely morphology-internal, should not be readily accessed by the syntactic component to condition the grammaticality of the form of the adjective. In a non-plural approach, the evident connection with plurality entailed in the homophony condition here appears especially

[5] In Russian, the impersonal form (i.e., 3rd-person neuter singular) of the finite predicate verb is used whenever subject–predicate agreement fails, due to lack of a subject, demotion of the subject, or otherwise.

thorny without resorting to the plural number. To my knowledge, all previous theories fail the mountain puzzle.

The crux of the puzzle lies in the feminine head noun's "double status as GEN.SG/NOM.PL" in the direct paucal when modified by a NOM.PL adjective, as in (5a). From (5e) we can see the head noun is GEN.SG, not NOM.PL, but for a NOM.PL adjective here to be acceptable in the 20th c. we need the head noun to be disguisable as NOM.PL, as the ungrammaticality of (5c) in the 20th c. suggests. How can these two apparently contradictory observations be reconciled in formal terms?

2.6 Interim summary

I have shown above that nonagreeing plural accounts of the Russian QNP are incompatible with the Ivanov(a) puzzle and the mountain puzzle, and agreeing nonplural accounts of the Russian QNP are incompatible with Suprun's puzzle and the mountain puzzle. Here's what we have learned from these puzzles:

- The Ivanov(a) puzzle: the solution to the nominal–adjectival number "nonagreement" doesn't lie in the syntax.
- Suprun's puzzle: the difference between the two adjective variants in the direct paucal is between a "more plural" and a "less plural" variant.
- The mountain puzzle: the "more plural" variant of the direct paucal is associated with "plural" and "paucal" at the same time.

Therefore, a new approach to the Russian QNP is called for to address these puzzles properly.

3 Plural, paucal, or both: an agreeing approach

In this section, I propose a unified approach to both quantified and unquantified NPs in Russian. This new approach is capable of solving all the above-mentioned puzzles, by allowing all nodes of a (Q)NP at the morphology–syntax interface to be uniformly labeled:

- SG = [+sg] (the quantified/unquantified singular number),
- PLX = [−sg, −pc, +pl] (the unquantified plural number; "plural exclusive"),
- PCPL = [−sg, +pc, +pl] (the paucal number sharing more features with the plural; "paucal plural"),
- PCX = [−sg, +pc, −pl] (the paucal number sharing fewer features with the plural; "paucal exclusive"), or
- ML = [−sg, −pc, −pl] (the multal number).[6]

The mechanism consists of two components:

- The morphological component builds paucal and multal cells of declensional paradigms by substantially employing rules of referral. Wordforms labeled with a bundle of features are presented to the morphology–syntax interface for feature checking.
- The syntactic component resolves a uniform morphosyntactic label across the (Q)NP from the semantics and the lexical classes of the numeral and the noun. This label, along with the semantics, serves as a starting point for determining the labels of external agreeing nodes (including peripheral adjuncts, and also the predicate verb if governed by a subject headed by this (Q)NP).

3.1 Morphological component: generating paradigms

Like previous paucal approaches, the nominal–adjectival split in the direct paucal is naturally borne out by the discrepancies between inflectional paradigms.

An adjectival (adjectives, and nouns converted therefrom) paradigm has a two-way nonsingular contrast in the category of number, maintained in direct cases and neutralized in oblique cases, as shown in (6). This two-way contrast may be identified as [+pl] (= PLX | PCPL) and [−pl] (= PCX | ML). As a rule of referral, the direct [−pl] is syncretized to GEN.PL.

[6] [−sg] is henceforth omitted in the presence of either [pl] or [pc] for the sake of brevity, as [+sg] is incompatible with them.

(6)

a. *bol'šój* "large" (Adj)

	SG.F	SG.N	SG.M	[+pl]	[−pl]
NOM	*bol'šája*	*bol'šóe*	*bol'šój*	***bol'šíe***	***bol'šíx***
ACC.INAN	*bol'šúju*				
ACC.ANIM			*bol'šógo*	***bol'šíx***	
GEN	*bol'šój*				
PREP			*bol'šóm*		
DAT			*bol'šómu*	*bol'ším*	
INS	*bol'šój(u)*		*bol'ším*	*bol'šími*	

b. *stolóvaja* "canteen" (Noun.F.INAN)

	SG.F	[+pl]	[−pl]
NOM	*stolóvaja*	***stolóvye***	***stolóvyx***
ACC	*stolóvuju*		
GEN	*stolóvoj*	***stolóvyx***	
PREP			
DAT		*stolóvym*	
INS	*stolóvoj(u)*	*stolóvymi*	

A nominal paradigm (nouns excluding those converted from adjectives) has an at most three-way non-singular contrast in the category of number, again maintained in direct cases and neutralized in oblique cases, as shown in (7). This three-way contrast may be identified as PLX, [+pc] (= PCPL | PCX), and ML. As rules of referral, DIR.ML is syncretized to GEN.PL, and DIR.[+pc] is typically syncretized to GEN.SG[7]—but in the case of a pl. tantum, DIR.[+pc] is defective, because no GEN.SG is available within its paradigm.

(7)

a. *gorá* "mountain" (Noun.F.INAN)

	SG	PLX	[+pc]	ML
NOM	*gorá*	*góry*	***gorý***	***gór***
ACC	*góru*			
GEN	***gorý***	***gór***		
PREP	*goré*	*goráx*		
DAT		*gorám*		
INS	*gorój*	*gorámi*		

b. *kníga* "book" (Noun.F.INAN)

	SG	PLX	[+pc]	ML
NOM	*kníga*	*knígi*	***knígi***	***kníg***
ACC	*knígu*			
GEN	***knígi***	***kníg***		
PREP	*knígе*	*knígax*		
DAT		*knígam*		
INS	*knígoj*	*knígami*		

c. *časý* "clock" (Noun.PL-ONLY.INAN)

	SG	PLX	[+pc]	ML
NOM	—	*časý*	—	***časóv***
ACC				
GEN	—	***časóv***		
PREP	—	*časáx*		
DAT	—	*časám*		
INS	—	*časámi*		

d. *párk* "park" (Noun.M.INAN)

	SG	PLX	[+pc]	ML
NOM	*párk*	*párki*	***párka***	***párkov***
ACC				
GEN	***párka***	***párkov***		
PREP	*párke*	*párkax*		
DAT	*párku*	*párkam*		
INS	*párkom*	*párkami*		

The numeral, the noun, and any adjective dependent on the noun of a QNP share the same morphosyntactic label resolved by some syntactic derivations to be specified later (see (12)), showing a completely agreeing pattern. The head noun *stolovaja* ("canteen") in (8ab) and the dependent adjective *stolovyj* ("related to dinner") in (8cd) share a range of inflected forms, as since the adjective's paradigm is a subset of the noun's; the head noun *stolovaja* in (8ab) can take a form syncretized to GEN.PL (*stolovyx*) but the head noun *komnata* ("room") in (8cd) can't (GEN.PL *komnat*), simply because the former belongs to the adjectival declension and the latter to the nominal declension.

(8) "two large canteens" (NOM)

a.	*dve*	*bol'šie*	*stolovye*	
	two.NOM.F	large.PCPL.NOM	canteen(Noun.F).PCPL.NOM	
b.	*dve*	*bol'šix*	*stolovyx*	
	two.NOM.F	large.PCX.NOM	canteen(Noun.F).PCX.NOM	
c.	*dve*	*bol'šie*	*stolovye*	*komnaty*
	two.NOM.F	large.PCPL.NOM	related_to_dinner(Adj).PCPL.NOM	room(Noun.F).[+pc].NOM

[7] Unless there's a lexically specified form (traditionally called *счётная форма*, lit. "counting form") in parallel with the syncretized form, which is the case for 5 exceptional masculine nouns of the 2nd declension: for *čás* "hour", DIR.[+pc] = *časá* (counting form), *časa* (= GEN.SG); etc. The syncretized form can be used whenever emphasizing individuality; the unsyncretized form can be used whenever the numeral and the head noun are adjacent. See Mel'čuk (1985: 430+) for more on the condition.

d. *dve* *bol'šix* *stolovyx* *komnaty*
two.NOM.F large.PCX.NOM related_to_dinner(Adj).PCX.NOM room(Noun.F).[+pc].NOM

3.1.1 The Ivanov(a) puzzle revisited

Now let's return to the Ivanov(a) puzzle. The paucal and multal inflections of this mixed declension are just natural extensions of the unquantified side (singular and plural). To anticipate, the syntactic component licenses at the morphology–syntax interface PCX (= [+pc, −pl]) for referents of any single or mixed gender, and PCPL (= [+pc, +pl]) for only all-female referents.

- In the adjectival subparadigm (see (9a)), the direct PCPL is naturally syncretized to the direct PLX (*Ivanovy*). This generates nominative *tri Ivanovy* ("3 Ivanovas", all female), which is compatible with the all-female label PCPL at the interface.
- In the adjectival subparadigm (see (9a)), the direct [−pl] (= PCX | ML) is syncretized to GEN.PL (*Ivanovyx*). This generates nominative *tri Ivanovyx* ("3 Ivanov(a)s", any single or mixed gender), which is compatible with the gender-neutral label PCX at the interface.
- In the nominal subparadigm (see (9b)), the direct [+pc] (= PCPL | PCX) is syncretized to GEN.SG, namely only GEN.SG.M (*Ivanova*), but not GEN.SG.F (**Ivanovoj*) as it is not in the nominal subparadigm. This generates nominative *tri Ivanova* ("3 Ivanovs", all male), which is compatible with the gender-neutral label PCX at the interface.

(9) a. Adjectival subparadigm of *Ivanov(a)* (**extrapolated forms*)

	SG.F	SG.M	PLX | PCPL	PCX | ML
NOM	*Ivanóva*	*Ivanóv*	***Ivanóvy***	***Ivanóvyx***
ACC.INAN				
ACC.ANIM	*Ivanóvu*			
GEN	*Ivanóvoj*	**Ivanóvogo*	***Ivanóvyx***	
PREP		**Ivanóvom*		
DAT		**Ivanóvomu*	*Ivanóvym*	
INS	*Ivanóvoj(u)*	*Ivanóvym*	*Ivanóvymi*	

b. Nominal subparadigm of *Ivanov(a)* (**extrapolated forms*)

	SG.F	SG.M	PLX	[+pc].M	[+pc].F	ML
NOM	*Ivanóva*	*Ivanóv*	*Ivanóvy*	***Ivanóva***	**Ivanóvy*	**Ivanóv(ov)*
ACC.INAN						
ACC.ANIM	*Ivanóvu*					
GEN	**Ivanóvy*	***Ivanóva***	**Ivanóv(ov)*			
PREP	**Ivanóve*	*Ivanóve*	**Ivanóvax*			
DAT		*Ivanóvu*	**Ivanóvam*			
INS	**Ivanóvoj*	**Ivanóvom*	**Ivanóvami*			

c. Conflated paradigm of *Ivanov(a)*

	SG.F	SG.M	PLX	PCPL	[+pc].M	PCX | ML
NOM	*Ivanóva*	*Ivanóv*	*Ivanóvy*	***Ivanóvy***	***Ivanóva***	***Ivanóvyx***
ACC.INAN[8]						
ACC.ANIM	*Ivanóvu*					
GEN	*Ivanóvoj*	***Ivanóva***	***Ivanóvyx***			
PREP		*Ivanóve*				
DAT		*Ivanóvu*	*Ivanóvym*			
INS	*Ivanóvoj(u)*	*Ivanóvym*	*Ivanóvymi*			

These forms are the only combinations available for *Ivanov(a)* in the direct paucal/multal within the range of variation,

[8] *Ivanov(a)*, being lexically animate, is surficially inanimate when quantified, say, by "22" (higher paucal) or "5" (multal).

and the resulting paradigm conflated from the adjectival and nominal subparadigms is given in (9c). Case closed.

3.2 Syntactic component: resolving morphosyntactic labels

The morphosyntactic labels accessed by the morphological component, are resolved by the syntactic component from the lexical classes of the numeral and the noun, and the intended semantics. This resolution process is by no means peculiar to the present approach; we have been familiar with some applications of this process, for example, in resolving the singular numeral–pl. tantum conflict, whatever approach or framework is to be adopted.

(10) a. *odin* *park* "**1** park" (NOM)
one(**SG**).**SG**.M.NOM park(M).**SG**.NOM
b. *dvadcat' odin* *park* "**21** parks" (NOM)
twenty-one(**SG**).**SG**.M.NOM park(M).**SG**.NOM
c. *odni* *časy* "**1** clock" (NOM)
one(**SG**).**PL**.NOM clock(**PL-ONLY**).**PL**.NOM

In Russian, "1" and "21" are lexically singular numerals, although "21" is semantically plural. When "1" is in combination with an ordinary noun, there's no conflict in number and the phrase takes a surface singular number, as in (10a). When "21" is in combination with an ordinary noun, the intrinsic singularity of the numeral overrides the semantic plurality and the phrase takes a surface singular number, as in (10b). However, when "1" is in combination with a pl. tantum, the intrinsic plurality of the noun takes priority over both the intrinsic singularity of the numeral and the semantic singularity, thus yielding a surface plural number, as in (10c).

When the semantic number, the plural-onliness of the noun, and the lexical number of the numeral as the inputs to the resolution process are listed against the output labels, we get (11):

(11) Labels at the morphology–syntax interface v. numeral class, noun gender, and semantic number

Numeral example	5	2, 22	21	1	(unquantified)	
Numeral class	multal	paucal	singular	singular		
Semantic number	plural	plural	plural	singular	singular	plural
Noun gender: F	ML	[+pc]	SG	SG	SG	PLX
Noun gender: M/N	ML	[+pc, −pl]	SG	SG	SG	PLX
Noun gender: PL-ONLY	ML	[+pc, −pl][9]	PLX	PLX	PLX	PLX

From (11) we can deduce the following resolution process:

(12) a. If the numeral is lexically multal (like "5") or paucal (like "2" or "22"), then label the QNP as ML or [+pc] respectively.
i. In addition, a [+pc] QNP is normally[10] labeled [+pc, −pl] (= PCX) if the noun is not feminine.
b. Otherwise, if the noun is plural-only, then label the QNP as PLX.
c. Otherwise, if the numeral is lexically singular (like "1" or "21"), then label the QNP as SG.
d. Otherwise, label the QNP as SG or PLX according to the semantic number.

Thus, the following labels can be derived (with *kniga* "book" being feminine, *park* "park" being masculine, *časy* "clock" being plural-only):

(13) a. "5 large books/parks/clocks": ML per Rule (12a)
b. "2 large books": [+pc] (= PCPL | PCX) per Rule (12a)

[9] Pl. tantum are actually indifferent to being specified as either [+pc] or [+pc, −pl], as they are ineffable in the nominative paucal due to their defectiveness in [+pc]. The choice of the label here and the resulting formulation in (12a.i) are arbitrary.

[10] Here a [+pc, +pl] adjective (syncretized to NOM.PL) is not at all impossible with a masculine/neuter noun today but much more restricted in terms of its condition (see Mel'čuk 1985: 176), and was in competition with the [+pc, −pl] variant (syncretized to GEN.PL) until the 19th century (see Madariaga & Igartua 2017: 113). Put otherwise, this rule has arisen as the result of the decline of the [+pc, +pl] adjective with a masculine/neuter noun in the 20th century.

c. “2 large parks/clocks”: PCX per Rule (12a.i)
d. “{a, 1} large clock”, “(21) large clocks”: PLX per Rule (12b)
e. “1 large book/park”, “21 large books/parks”: SG per Rule (12c)
f. “a large book/park”: SG per Rule (12d)
g. “large books/parks”: PLX per Rule (12d)

Hence the appropriate forms in the nominative case:

(14)				
a.	“5 large books/clocks” (NOM) ML licensed by Rule (12a)	*pjat’* five.NOM	*bol’šix* large.ML.NOM	*knig/časov* book(F)/clock(PL-ONLY).ML.NOM
b.	“2 large books” (NOM) PCPL licensed by Rule (12a)	*dve* two.NOM.F	*bol’šie* large.PCPL.NOM	*knigi* book(F).PCPL.NOM
c.	“2 large books” (NOM) PCX licensed by Rule (12a)	*dve* two.NOM.F	*bol’šix* large.PCX.NOM	*knigi* book(F).PCX.NOM
d.	“2 large parks” (NOM) PCX licensed by Rule (12a.i)	*dva* two.NOM.M/N	*bol’šix* large.PCX.NOM	*parka* park(M).PCX.NOM
e.	“2 large clocks” (NOM) PCX licensed by Rule (12a.i)	(ineffable[11] because *časy* “clock” is defective in DIR.PCX)		

3.2.1 Predicate agreement and Suprun’s puzzle revisited

We can now directly derive predicate agreement with only this uniform label of the QNP and the semantics in hand, without further referring back to the lexical classes of the numeral and the noun. We can list the range of all possible verb agreement options against lexical classes and the semantics as follows:

(15)

Numeral example	5	2, 22	21	1	(unquantified)	
Numeral class	multal	paucal	singular	singular		
Semantic number	**plural**	**plural**	**plural**	singular	singular	**plural**
Noun gender: F	ML	[+pc, **+pl**]	SG			[−pc, **+pl**]
Noun gender: M/N/F		[+pc, −pl]				
Noun gender: PL-ONLY		(ineffable)				
Noun gender: F	**0/PL**	**PL**	**SG.=/PL**	SG.=		**PL**
Noun gender: M/N/F						
Noun gender: PL-ONLY		(ineffable)				

Subject QNP

Predicate verb

These agreeing options can thus be derived in the following way:

(16) a. If the subject QNP is [+pl], plural agreement (PL) of the verb is the only option (shaded blue in (15)).
b. Otherwise,
i. singular agreement with agreeing gender (SG.=) is an option if the subject QNP is SG;
ii. neuter singular nonagreement (0) is an option if the subject QNP is [+pc] or ML;
iii. plural agreement (PL) is an option if the semantics of the subject QNP is plural (shaded red in (15)).

Now let’s return to Suprun’s puzzle, which can be readily accounted for with the following glossing:

(17) “On the table lay four large books.”

a.	✗	*Na stole*	*ležalo*	*četyre bol’šie knigi.*
		on the table	lie.PST.**SG.N**	four large books(F).[+pc, **+pl**].NOM
b.	✓	”	*ležali*	*četyre bol’šie knigi.*
		”	lie.PST.**PL**	four large books(F).[+pc, **+pl**].NOM
c.	✓	”	*ležalo*	*četyre bol’šix knigi.*
		”	lie.PST.**SG.N**	four large books(F).[**+pc**, −pl].NOM
d.	✓	”	*ležali*	*četyre bol’šix knigi.*

[11] Yet effable using the collective construction that doesn’t require this form: *dvoe* (collective) *bol’šix* (NOM.ML) *časov* (NOM.ML).

” lie.PST.**PL** **four** large books(F).[+pc, –pl].NOM

Labeled [+pc, +pl], *četyre bol'šie knigi* allows only plural agreement of the verb as in (17b) per Rule (16a), but not nonagreement as in (17a). Labeled [+pc, –pl], *četyre bol'šix knigi* allows neuter singular nonagreement as in (17c) per Rule (16b.ii), or semantic plural agreement as in (17b) per Rule (16b.iii). Case closed.

3.2.2 The mountain puzzle revisited

This also provides a straightforward explanation for the mountain puzzle in the new approach, where in the direct paucal a NOM.PL adjective is associated with PCPL, and a GEN.PL adjective with PCX. Thus, the apparent "double status" of the head noun is captured by the [+pc, +pl] nature of the PCPL number, and the ungrammaticality of a feminine noun in combination with a NOM.PL adjective in the direct paucal is a mere reflection of the defectiveness of the DIR.PCPL cell in the noun's paradigm:

(18) 20th-c. Russian (cp. (7), 21st-c. Russian)

a. *gorá* "mountain" (Noun.F.INAN)

	SG	PLX	PCPL	PCX	ML
NOM	*gorá*	*góry*	—	***gorý***	***gór***
ACC	*góru*				
GEN	***gorý***	***gór***			
PREP	*goré*	*goráx*			
DAT		*gorám*			
INS	*gorój*	*gorámi*			

b. *kníga* "book" (Noun.F.INAN) (= (7b), 21st-c. Russian)

	SG	PLX	PCPL	PCX	ML
NOM	*kníga*	*knígi*	***knígi***	***knígi***	***kníg***
ACC	*knígu*				
GEN	***knígi***	***kníg***			
PREP	*knígе*	*knígax*			
DAT		*knígam*			
INS	*knígoj*	*knígami*			

The defectiveness of DIR.PCPL for *gora* in 20th-c. Russian is well-motivated: this cell shares part of its number feature with PLX and part with PCX, and attempts to be identical with both at the same time. This works fine for most feminine nouns (like *kniga* "book") which have identical wordforms in the three cells in question (*knígi* = DIR.PLX = DIR.PCPL = DIR.PCX), but the same doesn't hold for words like *gora*, where there is a split in stress between DIR.PLX (*góry*) and DIR.PCX (*gorý*). When this is the case, DIR.PCPL would rather be left defective so as not to instigate a conflict.

The decline of the mountain rule is also straightforward: with the use of the DIR.PCPL variant of adjectives on the rise during the 20th century in the direct paucal with most feminine nouns like *kniga* (Nesset 2020), analogical expressions started to be used with words like *gora* as well, breaking the original ineffability. In this way, the DIR.PCX form of nouns like *gora* extended to DIR.PCPL, filling the paradigmatic gap.

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